Sandip Dayani

Contact

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Skills

- Material Modeling and Simulation
- Crystal Plasticity and Damage Mechanics
- Experimental Material Characterization
- Multi-physics Simulations
- Finite Element Analysis (LS-Dyna, ANSYS, Abaqus)
- Python, Fortran, MATLAB
- CAD Tools (Inventor, Creo, CATIA)
- Teaching and Academic Support
- Technical report writing and presentation

Research Experience

- Developed damage evolution model and implemented it into a crystal plasticity framework at University of Waterloo (2022–Present)
- Simulated void nucleation and growth under high-strain-rate conditions using advanced kinetics models
- Conducted experimental tests on HSLA, DP and Q&P steels to analyze deformation behavior
- Built numerical models of borehole heat exchangers with PCM using ANSYS and validated with experiments (Fraunhofer UMSICHT, 2020)
- Simulated laser metal deposition processes and optimized shielding design in additive manufacturing (Fraunhofer IWS, 2020)
- Analyzed thermal distribution in carbon fiber manufacturing using FEA tool (DLR, 2019)
- Modeled non-linear elastic behavior of fiber-reinforced composites using RVE methods (TU Chemnitz, 2019)

PhD candidate in Mechanical and Mechatronics Engineering at the University of Waterloo with a strong focus on fracture mechanics, damage evolution modeling, and finite element simulations. Proven experience in both experimental characterization and numerical modeling of high-strength steels under dynamic loading, with publications in leading journals. Skilled in structural integrity assessments, fatigue and crack growth analysis, and application of deterministic and probabilistic methods for component life evaluation. Proficient with ANSYS, Abaqus, LS-DYNA, and programming in Python, Fortran, and MATLAB. Experienced in developing and validating models for material degradation and crack propagation. A detail-oriented and analytical engineer, committed to supporting safe, efficient, and innovative solutions in the energy and nuclear sectors. Brings international research experience from Fraunhofer, DLR, and TU Chemnitz, with a passion for applying advanced mechanics to solve real-world engineering challenges.

EDUCATION

Expected in Jan 2026

Doctor of Philosophy (PhD) in Mechanical and Mechatronic Engineering

University of Waterloo, Waterloo, Canada

- GPA: 3.7

May 2021

Master of Science (M.Sc.) in Merge technologies for resource efficiency

Technische Universität Chemnitz, Chemnitz, Germany

- GPA: 3.5

June 2017

Bachelor of Engineering (B.E) in Mechanical Engineering

Gujarat Technological University (GTU), Shree Swami Atmanand Saraswati Institute of Technology, Gujarat, India

- GPA: 3.5

Work History

Jan 2022 – Current

Doctoral Research Assistant, **University of Waterloo**, Waterloo, Canada

- Designed and implemented a thermodynamically consistent damage evolution model into a crystal plasticity framework using Fortran, tailored for advanced structural integrity analysis.
- Investigated void nucleation and crack growth behaviour under high-strain-rate loading, supporting predictions for fracture-critical components.
- Conducted mechanical testing on HSLA, DP, and Q&P steels and correlated results with simulation outcomes to validate structural reliability.
- Developed a physics-based model to estimate internal heat generation during plastic deformation, relevant to safety assessments under extreme loading conditions.

Publications

- S. Dayani, W. Muhammad, A. Brahme, F. Hekmat, T. Sabiston, K. Inal, 2025. Thermodynamically consistent damage evolution model coupled with rate-dependent crystal plasticity: Application to high-strength low alloy steel at various strain rates. International Journal of Plasticity 186, 104255. https://doi.org/10.1016/j.ijplas.2025.104255

Certifications

- Certification courses of AutoCAD 2015 and Creo 2.0 from Authorized CAD Training Center, India
- Master Solidworks 2018 - 3D CAD using real-world examples (July 2021) : Certified from Udemy
- ANSYS Tutorial (Sep 2019): Certificate from Udemy
- Mastering ANSYS CFD (May 2020): Certificate from Udemy
- Industrial Catia V5 R20 (Dec 2019): Certificate from Udemy
- Learning Autodesk Inventor 2016(Jul 2020): Certificate from Udemy

Languages

- English: Fluent
- German: Intermediate
- Gujarati: Mother tongue
- Hindi: Native

- Utilized LS-DYNA, Abaqus, and in-house tools to simulate fracture and fatigue behavior for safety-critical materials.
- Authored peer-reviewed research in high-impact journals, focusing on fracture mechanics, damage modeling, and probabilistic material response.

Sept. 2022 – Dec. 2022, Sept. 2024 – Dec. 2024
Teaching Assistants, **University of Waterloo**, Waterloo, Canada
ME559, ME219

- Facilitate and lead educational tutorials, providing guidance and support
- Evaluate and grade student assignments
- Assist students in the development and completion of their projects
- Provide individualized support to students during designated office hours

Jul 2020 – Apr 2021
Master Thesis, **Fraunhofer Gesellschaft (UMSICHT),** Oberhausen, Germany

➢ Modelling and simulation of the different borehole heat exchanger with PCM as backfill material

- Developed FEA models in ANSYS for simulating borehole heat exchangers and validated against experimental data to improve component life under cyclic thermal loading.
- Performed parametric sensitivity analysis and thermal simulations for structural optimization using PCM-enhanced materials.
- Results contributed to improving design reliability in thermally loaded systems.

Jul 2020 – Mar 2021
Research Assistant, **Fraunhofer Gesellschaft (IWS),** Dresden, Germany

➢ Adaption and construction of a shielding nozzle for COAXshield

- Led simulation-based redesign of COAXshield using ANSYS CFX, optimizing flow performance under thermal and structural loads.
- Applied FE-based parametric optimization to enhance shielding device performance in additive manufacturing processes.
- Integrated CAD and FEA tools to validate structural stability and thermal effects under operational scenarios.

Sept 2019 – Dec 2019
Intern, **German Aerospace Centre (DLR),** Augsburg, Germany

➢ Thermal simulation of the automated fibre placement process with FEA

- Conducted thermal-structural simulations using ANSYS to model the automated fibre placement process, focusing on high-temperature behaviour of composite components.
- Defined boundary conditions and loading to predict thermal gradients and induced stress fields, relevant for long-term structural integrity of aerospace parts.
- Compared simulation data with experimental results to validate the modelling accuracy and support process optimization.

Apr 2019 – Aug 2019
Research Project Assistant, **Technische Universität Chemnitz,** Germany

➢ Micromechanical modelling of fibre-reinforced plastic with non-linear elastic behaviour

- Developed micromechanical FEA models (RVE) in ANSYS to calculate material constants of fibre-reinforced composites, including non-linear behaviour.
- Created material maps for input into macro-scale simulations, aiding multi-scale integrity assessments of structural materials.
- Supported model validation against theoretical predictions and classical mixture rules.